

Liars Are Information: Threshold-Gated Inversion of Coherent Adversaries in Multi-Agent LLM Swarms, and Its Limit

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Abstract—Byzantine fault tolerance for multi-agent systems is built on discarding, and the resulting 50% ceiling is a property of discarding rather than of the problem. Correlated error is known to cap evaluation panels; we confirm that ceiling in an adversarial swarm and extend it to the replicated-model case a deployment actually builds (0.665 within-model against 0.432 cross-family), with a calibrated law rather than an observation. Over all 35 three-model subsets of our roster no composition beats its own strongest single member by a detectable margin, and on MATH-500 every subset is worse by at least -0.140 : an LLM swarm’s ceiling on honest error is *correlation*, not *competence*. The only structure in the broadcast set that honest agents do not already share is therefore *adversarial* coherence: a disciplined liar is a swarm’s one informative contributor, and can be inverted rather than filtered.

The obstacle is that honest models also agree on wrong answers (0.547 pooled across powered four-way multiple-choice pairs, against chance 0.333), so we gate inversion on a per-answer-space calibrated threshold estimated blind via a multiclass attenuation law exact at $C = 2$. Across 7 attack classes and 6 benchmarks with prompt-injected adversaries, gated inversion is worth 0.645 at $f = 0.7$ and costs 0.001 — below the floor — at $f = 0$.

We then defeat our own mechanism. An adversary given the published calibration constant coordinates just beneath it, driving accuracy to 0.082 at the exact tie; its optimum coherence, 0.280, lies *below* the chance value 0.333 at which honest agents collide by accident. No rule measurable with respect to per-question coherence separates the two. One finding outlives the mechanism: the logprob confidence surface discriminates correctness better than the self-report (0.704 against 0.628) and cannot be authored by the sender, whose falsified self-report drops below chance to 0.372. Ten claims this study set out to confirm were refuted by its own data, and are reported at the same length as the confirmations.

Index Terms—Byzantine fault tolerance, multi-agent systems, large language models, error correlation, adversarial robustness

I. INTRODUCTION

A decentralized swarm of language-model agents answers a question by broadcasting individual answers and locally aggregating what it hears. When a fraction f of the agents is Byzantine, the aggregation rule decides whether the swarm survives, and essentially every rule in the literature survives by *discarding*: Krum selects the contribution closest to its neighbours [1], coordinate-wise median and trimmed

mean suppress outlying coordinates [2], confidence weighting down-weights low-confidence peers, filter-and-refine schemes drop contributors whose history looks unreliable. The shared premise is that adversarial input is contamination, and it yields the familiar ceiling: once adversaries hold the plurality, a rule that treats consensus as evidence follows them.

A. Why a swarm needs an adversary to be worth pooling

Before defending a swarm it is worth asking what pooling honest agents buys, and the answer we measure is: very little. Correlated error already caps evaluation panels [3]–[5]; we confirm that ceiling in an adversarial swarm and extend it to the case a deployment actually builds, replicating one model, where error correlation is 0.665 against 0.432 across families. Over every three-model subset of our roster no composition beats its own strongest single member by a detectable margin, and maximising effective swarm size correlates with swarm accuracy at -0.506 — the wrong sign.

An LLM swarm’s ceiling on honest error is correlation, not competence. The consequence is sharp: the only structure in the broadcast set that honest agents do not already share is *adversarial* coherence. A disciplined liar is the one genuinely informative contributor a swarm has.

B. Liars are information

An adversary that lies *coherently* — reporting the same wrong answer as its collaborators, task after task — is not injecting noise. It is operating a channel whose sign is recoverable: if a bloc reliably says X when the truth is not X , then “this bloc says X ” is evidence *against* X . Discarding it throws that away; inverting it converts the adversary into a contributor.

Measuring coherence requires no labels: with e_i an agent’s error rate and q_{ij} the probability that two agents who are both wrong are wrong *in the same way*, agreement obeys $m_{ij} = (1 - e_i)(1 - e_j) + e_i e_j q_{ij}$, of which the binary case used in prior work is the instance $q_{ij} \equiv 1$. The coincidence term is the inversion signal, not a nuisance parameter.

The obstacle is that honest agents are not idiosyncratic either — pooled across the powered four-way multiple-choice

pairs, two honest agents that are both wrong pick the *same* wrong answer 0.547 of the time, against a 0.333 chance level — so a rule triggered by coherence alone inverts honest agents precisely on the questions they find hardest. Our gate therefore fires only when a channel’s coherence exceeds a calibrated honest ceiling *and* it agrees with the receiver less than the attenuation law permits. Both tests are anchored on the receiver’s own answer, which is what keeps them meaningful past $f = 0.5$.

C. The paper’s turn

The mechanism works, and then it is defeated. An adversary that knows the gate exists does not have to beat the threshold — it can stay under it. We construct one, show it reduces AIP to 0.082 at the exact tie, and prove that no rule computed from per-question coherence separates it from honest independent disagreement, because at its optimum it is *less* coherent than honest agents are. Of the pre-registered mitigations, soft gating partly works and threshold randomisation does nothing measurable.

D. Contributions

- **Mechanism.** A local three-way channel decision (invert / trust / discard) on a calibrated honest-coincidence threshold, with a windowed variant for non-stationary adversaries.
- **Theory.** A multiclass attenuation law with its identifiability conditions; *coordinatability*; and Proposition 1, a total-variation limit showing a residual attack at or below chance coherence is invisible to *every* rule measurable with respect to per-question coherence.
- **Design laws: confirming and extending the correlation ceiling.** The ceiling is established for evaluation panels [3]–[5]; we confirm it under adversarial conditions and extend it three ways — to replicated models, from an observation to a law estimable blind at run time, and from a measurement caveat to the premise of a defence.
- **Protocol.** Of the two confidence surfaces an agent can broadcast, the logprob-derived one discriminates better (AUC 0.704 against 0.628) and cannot be authored by the sender, whose falsified self-report drops to 0.372, below chance. This is the one recommendation here the gate-aware adversary does not reach.
- **Evaluation, including what failed.** 7 attack classes, a pre-registered mitigation study with criteria committed before the runs, and a cost table showing the gate is 3.6% of the aggregation cost it belongs to. Ten claims this study set out to confirm were refuted by its own data and are reported at the same length as the confirmations.

Every effect is compared against a resampling floor: 0.066 on GSM8K, 0.099 on the 200-task benchmarks. An effect below its cell’s floor is **undetectable at this scale**, never “small” or “no effect”.

II. RELATED WORK

We organise prior work by what it does with a suspicious contribution: *discard* it, *model* it, or *punish* it. Our mechanism

belongs to none of these, because it *uses* the contribution. Two further subsections cover the measurement result we build on and the theoretical result we complement.

A. The discard paradigm

Robust aggregation for distributed learning set the template: Krum selects the vector minimising summed distance to its neighbours [1], coordinate-wise median and trimmed mean achieve order-optimal rates under a bounded adversarial fraction [2], geometric median generalises the same intuition. All are designed for $f < 1/2$ and all assume the adversarial contribution carries no usable information. Within LLM multi-agent systems the template recurs at the response level, through confidence weighting [6], receiver-side re-evaluation and filtering [7], certifiable agreement instruments [8], learned aggregation [9], long-term incentives [10] and decentralized coordination [11], within a broader effort to make multi-LLM networks trustworthy [12] whose threat surface is itself an active subject [13].

Our results reproduce the expiry: against a symbolic coherent adversary at $f = 0.7$ every *deployable* discard baseline reads 0.000 on all 6 benchmarks. Two qualifications belong here rather than in a footnote, because both weaken the comparison in our favour if omitted. The collapse to exactly zero is a property of the symbolic adversary — against real prompt-injected adversaries nothing reaches zero, deployable discard spans 0.002 to 0.600, and on BoolQ plain majority vote *beats* AIP by -0.295 . And methods handed the true Byzantine count do not collapse even symbolically: Krum holds 0.815 on BoolQ. We exclude those as undeployable rather than as beaten, and say so wherever the comparison is made.

B. AgentChain and the identity assumption

AgentChain [14] is the closest system-level comparison and is unusually explicit about its boundary: its guarantees do not extend to $f \geq N/2$, and it names external identity verification as the unsolved problem. We cross that line without solving identity, by anchoring the gate on the receiver’s own answer rather than the plurality so the tests do not degrade when the plurality is adversarial — a different resource buying a different guarantee, one Section VII shows an adaptive adversary dissolves. When it dissolves, their gap is exactly where the argument lands. We do not close it; we give a reason to think a defence in our family *cannot*, which turns an engineering admission into a statement about the statistic.

C. The correlation ceiling in model panels

The result our measurement section builds on is not a defence. Kim et al. [3] establish that LLM errors are substantially correlated across models, and Kohli [4] shows what that costs a panel: nine judges are worth roughly two effective votes, so panel size overstates reliability. Jiang et al. [5] document the same homogeneity from the generation side.

We confirm this ceiling and extend it; we do not claim to discover it. *First*, the deployment-relevant comparison. Their panels are built from distinct models; a deployed swarm is

more likely built by replicating one, and that case is roughly half again as correlated (0.665 within-model against 0.432 cross-family). *Second*, from observation to calibrated law: the multiclass attenuation law is exact at $C = 2$ and validated against that exactness, which turns a descriptive finding into a constant an aggregator calibrates on at run time without labels. *Third*, correlation as premise rather than caveat: for a panel the ceiling is a measurement caveat, but for a swarm under attack it is the premise of the design — if honest pooling is capped, the only informative structure left is what honest agents do *not* share, which is adversarial coherence. Their negative result about panels is what makes our positive result about inversion necessary rather than merely available.

D. Detection theory: what a richer monitor buys

The closest theoretical neighbour points the opposite way, and the contrast is the most useful thing we can say about either. The Detection Dilemma [15] establishes in its Theorem 1 that a monitor observing *four* channels can force honesty: with enough independent views, misreporting becomes detectable and the rational strategy collapses to truthful reporting. Our Proposition 1 establishes that a rule measurable with respect to a *single* per-question coherence statistic is powerless against an adversary at or below chance coherence, because there the two hypotheses induce the same distribution.

These bracket the design space from opposite ends — an existence statement about monitors against an impossibility statement about one channel — and are complementary rather than competing: the question is not whether coherence-based gating can be repaired but how many independent observations a defence needs, and for the one-statistic case the answer is more than one. That is also why external information is the first escape class in Section VIII. We do not evaluate a four-channel monitor and make no claim about whether their bound survives our gate-aware adversary.

E. Modelling and punishment

Dawid–Skene [16] and its descendants treat unreliable contributors as parameters rather than points to reject. That is closer to our spirit, but it assumes conditional independence given the truth — exactly what a coordinated adversary violates — and estimates reliability rather than exploiting the systematic *direction* of error. We use a one-coin variant inside our blind estimator and as a baseline.

A third line disincentivises adversaries through staking, slashing or reputation decay, or analyses the strategic setting directly: a Byzantine agent’s broadcast is cheap talk, costless and not inherently credible [17]. This is orthogonal to aggregation and we make no claim in that direction. We must also distinguish work on premature convergence in multi-agent deliberation [18], which operates at *token-level generation* where shared context collapses diversity before an answer exists. We operate at *response-level aggregation* on a pathology that is adversarial rather than conformist; the two are complementary and their mitigations do not compete.

All inference here is executed offline and cached with vLLM [19]; the aggregation experiments never invoke a model (Section X).

III. SYSTEM, THREAT MODEL, AND THE GATE

A. Swarm and broadcast model

$N = 10$ agents answer the same task independently. Each broadcasts three things: an answer, a confidence derived from the mean token logprob over the answer span, and a separately elicited self-reported confidence. There is no central aggregator and no shared state; every agent computes `aggregate(observations, self)` locally, and the “swarm answer” we report is the answer a receiver holds after aggregating.

Topology is complete by default, with ring and k -nearest variants; partial observability $p_{obs} \in \{1, 0.5, 0.25, 0.10\}$ thins which broadcasts a receiver hears, and it rather than the Byzantine fraction is the binding practical constraint (Section IX). A receiver’s own vote is a confound for every weighted rule, so all comparisons run at matched self-vote share; the audit establishing that it is a real confound, and where it is not controlled, is in Section VI. [M2a, M2b]

B. Threat model

The Byzantine fraction ranges over $f \in \{0, \dots, 0.7\}$. Adversaries may collude, may know the defence in full, and in Section VII are given the published calibration constants; they cannot forge identities and cannot observe ground truth.

Three deployability constraints apply to every method we are willing to call a defence: no oracle knowledge of f , no labels at run time, and no trusted third party. Krum, Multi-Krum and trimmed mean are handed the true Byzantine count and are therefore marked *oracle* and reported separately. This is not a rhetorical exclusion. Those methods win on two benchmarks under the real-LLM adversary—multi-Krum reads 0.745 on MedQA where AIP reads less—and any claim that AIP dominates the discard family is a claim about the deployable subset only. [H2’]

C. Attack classes

7 attack classes are evaluated. Four are generated by prompt-injected LLM adversaries and require inference: `always_wrong` (assert a wrong answer), `semantic_negation` (a structured lie computed by a rule the bloc shares), `semantic_hallucination` (a fluent but idiosyncratic wrong answer), and `rushing` (condition on cached honest broadcasts and answer first).

Three are exact offline transforms of a cached adversarial run, which is what makes them clean rather than cheap. `falsified_confidence` inverts the self-reported confidence and leaves the answer and the logprob byte-identical, so any difference it produces is attributable to the self-report channel and to nothing else. `burst` schedules the coherent-lie cache against noise in alternating epochs. The adaptive bandit adversary chooses among arms across epochs according to realised payoff.

Invertibility tracks measured channel coherence, which is the paper’s central diagnostic. On GSM8K, negation reaches $\hat{q} = 0.674$ and is exploitable; rushing (0.156) and always_wrong (0.102) sit below the honest ceiling and are not; and semantic_hallucination (0.066) carries no channel at all despite being the most damaging attack in the set. An attack with no coherent channel is not exploitable however damaging it is. [I1, G2’]

The seventh class, an adaptive bandit adversary, has no fixed coherence: it chooses among the arms above across epochs.

D. The three-way channel decision

For each peer a receiver makes one of three decisions: INVERT, TRUST, or DISCARD.

INVERT requires *two* conditions. The channel’s coherence must be significantly above the calibrated honest ceiling for the answer space (one-sided exact binomial test against a derived minimum-support floor), *and* the channel must agree with the receiver less than the attenuation law permits for an honest peer of the estimated error rate. The first test alone is what a naive mechanism uses and it fails, because honest agents cohere. Both are anchored on the receiver’s own answer rather than the plurality, which is why the mechanism survives past $f = 0.5$: a plurality-anchored statistic has already converged on the lie there. [G1]

An inverted channel is not merely dropped. It contributes signed informativeness: weight $-w$ to the answer it broadcast and $+w/(K - 1)$ to each of the $K - 1$ alternatives, where w scales with the channel’s estimated capacity. This is the step that converts an adversary into a contributor, and the ablation in Section VI isolates its contribution by removing it.

E. Windowed statistics for non-stationary adversaries

Pooled channel history averages a burst adversary’s coherent and noisy regimes into a channel that clears neither threshold, so the pooled gate trusts every adversary. A sliding window of W tasks, together with a regime-change flag, restores the decision; a flagged channel is *demoted to discard* and never inverted, because a channel whose regime just changed is the one case where the estimated sign is least trustworthy. [B1]

F. Unfakeable confidence

The two confidence surfaces are cached separately for a reason. A prompt-injected agent can assert any self-reported confidence it likes; it cannot easily inflate the mean token logprob of a span it did not generate. AIP reads the logprob channel. Section V (Law 4) tests this directly: inverting the self-report moves AIP by exactly zero and moves a self-report-reading baseline by up to -0.163 . [F1]

IV. THEORY

A. The generalized attenuation law

Partition the event that two agents give the same answer by whether they are correct. Two agents can agree by both being right, or by both being wrong in the same way; a correct and an incorrect agent cannot agree, so the cross term vanishes

exactly. Writing e_i for agent i ’s marginal error rate and q_{ij} for the probability that two agents who are both wrong chose the same wrong answer,

$$m_{ij} = (1 - e_i)(1 - e_j) + e_i e_j q_{ij}.$$

The binary case is the special instance. With $C = 2$ there is exactly one wrong answer, so $q_{ij} \equiv 1$ identically and the law collapses to $2m_{ij} - 1 = (1 - 2e_i)(1 - 2e_j)$, the estimator used in prior work. This is not an approximation that happens to hold; it is an algebraic identity, and it gives us a control condition with no free parameters. On BoolQ, where $C = 2$, the measured q is 0.079 in mean absolute error against the multiclass law—exactness recovered from data, which is the strongest available evidence that the estimator is implemented as specified.

Applying the binary form when $q < 1$ overstates the quality product by exactly $e_i e_j (1 - q_{ij})$. The error is second order in the error rates, which is why it hides on easy benchmarks and surfaces on hard ones: on MATH-500 the binary law’s mean absolute error is 0.081 against the multiclass law’s 0.032. [C3, C4]

The chance level of q for a C -ary answer space is $q^{\text{chance}} = 1/(C - 1)$, and approximately zero for open-ended answers. This constant is the reference the whole paper turns on: it is what independent wrong answers look like, and Section VIII shows that an adversary sitting at it is indistinguishable from independent honest failure.

B. Identifiability and receiver conditioning

With q known — binary, at chance, or zero — the triplet identity recovers every e_i from $n \geq 3$ agents. With q_{ij} free, pairwise agreement gives $\binom{n}{2}$ equations for $n + \binom{n}{2}$ unknowns: short by exactly n for every n . This is structural unidentifiability, not a sampling problem, and no amount of data fixes it. Three escapes exist — a structural assumption on q , the full answer-pattern distribution as Dawid–Skene uses, or ground truth. We take the first at run time and use the third to check it. [C4]

There is also a gap between the quantity the theory names and the one a gate can compute, and closing it in the wrong direction is the easiest way to build a mechanism that validates and then fails. The law is stated in terms of $P(\text{same} \mid \text{both wrong})$, which needs labels; a receiver can only compute $P(\text{same} \mid \text{both dissent from me})$. The two coincide only when the receiver is almost always right, so they are different constants and the ceilings must be measured on the statistic the gate computes. Ours are receiver-conditioned throughout: honest coincidence is 0.535 on four-way multiple choice and 0.597 on open-numeric answers. [C3]

C. Coordinatability and the invertibility threshold

An adversary is exploitable only if it can be *coherent*, which requires a lie rule its members compute identically from the question alone — the property we call *coordinatability*. Which answer spaces permit it, and the direction in which that expectation failed, is measured in Section V (Law 3).

The threshold follows: invert only above the calibrated honest ceiling for the answer space. Section VIII asks what happens when an adversary reads that ceiling off the paper and sits underneath it.

V. SWARM-LEVEL DESIGN LAWS

Three properties of an honest swarm have to be measured before any defence is evaluated, because all three appear inside the mechanism as constants and all three constrain what a swarm can do at all. Each turns out to be a design law with a counter-intuitive edge: correlated failure is the default rather than the exception, decorrelation does not buy accuracy, and what an adversary can coordinate on is a property of the answer space rather than of the adversary. The first of the three is established prior art in the evaluation-panel setting [3]–[5]; we confirm it in an adversarial swarm and extend it, and we are explicit below about which parts are which.

Read together they make one claim: *an LLM swarm’s ceiling on honest error is correlation, not competence*, which leaves adversarial coherence as the only structure a swarm can exploit that its honest members do not already share.

A. Law 1: correlated failure is the default

This law is not ours to claim [3]–[5] (§II). What follows confirms the ceiling in an adversarial swarm rather than an evaluation panel, and extends it in two ways the adversarial setting forces.

The extension that matters for deployment is which pair is compared. A panel is assembled from distinct models; a deployed swarm is more likely built by replicating one, and replication is the more correlated case: error correlation between two agents running the *same* model at different sampling temperatures is 0.665 against 0.432 between different models, a factor of 1.54. The full pairwise matrix is in the artifact. Replicating a model does not replicate an independent opinion, and it is worse than mixing by more than half again. [C1]

The extension that matters for a defence is that the quantity becomes predictive. The multiclass attenuation law (Section IV) is exact at $C = 2$ and validated against that exactness, so the correlation structure is estimated blind at run time without labels — which is what lets it serve as the constant an inversion gate fires against.

In agent counts: a homogeneous swarm of 10 agents is worth 1.14–1.87 independent agents, mean 1.47; the best two-model mixture reaches 2.49. A ten-agent homogeneous swarm is closer to one epistemic agent than to ten — the swarm analogue of nine judges being worth two votes. [C2]

A basis restriction. Only GSM8K carries a measured within-model correlation, since only there did we sample the same model twice; elsewhere the mixed-swarm figures use cross-model correlation alone and are *upper bounds* on N_{eff} . The largest, 6.22, comes from a different estimator on an excluded benchmark and is not comparable.

TABLE I
COMPOSITION GUIDANCE. SWARM ACCURACY RANGE ACROSS ALL THREE-MODEL SUBSETS, AGAINST THE BENCHMARK’S MEASUREMENT FLOOR, AND THE BEST SUBSET’S GAIN OVER ITS OWN STRONGEST SINGLE MEMBER. COMPOSITION ONLY MOVES THE OUTCOME WHERE THE BENCHMARK LEAVES HEADROOM — AND NO COMPOSITION BEATS ITS BEST SINGLE MODEL BY A DETECTABLE MARGIN ANYWHERE.

Benchmark	range	floor	best gain over best single
MATH-500	0.170	0.099	−0.140
MedQA	0.285	0.099	+0.018
MMLU	0.103	0.099	+0.007
BoolQ	0.045	0.099	+0.015
ARC	0.043	0.099	+0.010
GSM8K	0.021	0.066	+0.013

B. Law 2: decorrelation does not buy accuracy

The obvious reading of Law 1 is that a practitioner should maximise N_{eff} . We tested that directly and it is wrong.

For every 35 three-model and 21 five-model subset we build a 10-agent swarm split as evenly as possible, average ρ over the real pair census, and compute both N_{eff} and plurality accuracy over the cached answers (Table I). Every distinct assignment of a surplus agent is enumerated and averaged, since the heavier model wins any three-way disagreement by weight alone.

N_{eff} turns out to be close to a restatement of roster weakness — Spearman −0.934 against mean roster accuracy, with every diversity-flavoured predictor losing its association once the mean is partialled out — and, more usefully, **it is negatively associated with swarm accuracy**: −0.506 at $k = 3$, against +0.658 for mean accuracy. Maximising effective swarm size selects *against* what a practitioner wants. What survives partialling is accuracy *spread*, partial Spearman about +0.7: among rosters of equal average strength, prefer unequal members — a second-order effect on top of “use accurate models”.

Family diversity cannot be assessed here: all 7 models come from 7 labs, so every k -subset contains exactly k lineages and the predictor is constant by construction — a limitation of the roster, not evidence that lineage is irrelevant.

Composition choice clears the floor on the three benchmarks with headroom and not on the three saturated ones; but on *no* benchmark does the best composition beat its own strongest single member detectably (largest +0.018), and on MATH-500 every three-model composition is *worse* than its best member by at least −0.140.

This is the negative result we would keep if we could keep only one. With Law 1 and the honest attractors it says an LLM swarm’s ceiling on honest error is **correlation, not competence**: the agents fail together, into the same wrong answers, and adding more does not help because what is added is not independent. We searched every three- and five-model subset of the roster for an exception and found none.

The consequence is direct: if honest pooling is capped, the only channel through which a swarm exceeds its best member is the structure honest agents do *not* share — adversarial coherence. That is what threshold-gated inversion spends,

which is why the positive result here is about robustness rather than accuracy, and why an adversary that declines to be coherent defeats the mechanism by giving it nothing to spend. [S1]

C. Law 3: coordinatability is a property of the answer space

An adversary is exploitable only if it can be *coherent*, which requires a lie rule its members compute identically from the question alone.

We expected coordinatability to collapse on closed label spaces, reasoning that four-way multiple choice offers no shared rule for choosing *which* wrong option to name. **The measurement runs the other way:** mean adversarial coherence is *highest* on the most closed space — 0.559 binary, 0.285 four-way multiple choice, against 0.104 open text. A smaller answer space makes accidental agreement easier, not harder.

What does collapse is the quantity a gate needs: the adversary’s *margin over honest agreement*. On four-way multiple choice the negation attack reaches $\hat{q} = 0.325$ against an honest receiver-conditioned floor of 0.535, and no attack’s interval clears that floor. On open-numeric answers the same attack reaches 0.674 against a floor of 0.597 and does clear it. On a closed space there is no coherent channel *relative to honest behaviour* for a gate to license inversion on. [G2’]

That floor is the gate’s *configured* constant rather than a measurement taken for the comparison, so it matters that the ordering survives either: the independently measured pooled value 0.547 exceeds it, which widens the margin the adversary fails to clear.

D. The constant the gate is calibrated on

The calibration constant is the rate at which two honest agents, both wrong, are wrong the *same* way. Pooled across the powered four-way multiple-choice pairs it is 0.547 against a chance level of 0.333; on MATH-500, where the answer space is effectively open, it is 0.027. Honest agents on a closed space are close to twice as likely to collide on a wrong answer as chance predicts, and that is the entire reason a coherence-triggered inversion rule needs a gate at all. [C3]

Power, and why the estimate is pooled. These are per-pair estimates conditioned on both agents being wrong, so the sample size is jointly-wrong items, not tasks. MATH-500 has all twenty-one pairs above the $n \geq 30$ threshold; the multiple-choice benchmarks have almost none individually (MedQA eight, MMLU one, ARC none), so the constant is pooled over the 19 powered pairs across them, 847 jointly-wrong items. An earlier version quoted the MMLU all-pairs mean, which rests on a single powered pair; it is withdrawn. Figure 3 draws underpowered pairs separately for the same reason.

E. Law 4: the broadcast contract decides what a receiver can believe

What an agent is obliged to broadcast is a security decision, not an interface detail. Every confidence-weighted defence in the literature reads a number the sender supplies; each

broadcast here carries two, the mean token logprob over the answer span and a number the model was asked to state. A prompt-injected agent can assert any self-report but cannot easily inflate the logprob of a span it did not generate.

As a predictor of its own correctness the logprob surface reaches ROC AUC 0.704 and the self-report 0.628, a gap of +0.076, over the 19 of 42 frozen-roster cells clearing the power floor (R2: at least thirty wrong answers). The weak-tier arm, reported separately, shows the same ordering.

The gap is not the argument — it is real but modest. The argument is what happens under attack. The *falsified_confidence* attack inverts the self-report and leaves answer and logprob byte-identical, so anything it changes is attributable to that channel. It drives self-report discrimination from 0.628 to 0.372 — *below chance*, on 18 of 19 cells. A receiver weighting on that surface is not merely uninformed, it is anti-informed. The logprob surface is unmoved exactly: AIP’s accuracy under this attack differs from its control by -0.001 on GSM8K and $+0.000$ on MMLU — an implementation identity rather than an estimate below the floor, since the gate never reads the self-report. [F1, P1]

A third failure mode needs no adversary: reasoning models frequently emit no parseable number on multiple choice, and the self-report is missing on up to 0.330 of rows in the worst powered cell. We report that rather than prompting until compliance improves, because the non-compliance is the finding — a field a well-behaved agent may omit and a misbehaving one may fabricate is a weaker commitment than one derived from tokens already emitted. **Broadcast both surfaces and weight on the logprob:** the one recommendation here that the gate-aware adversary does not reach, because it rests on no coherence statistic.

F. Blind estimator validation

The multiclass law beats the binary law wherever the answer space is not binary (MATH-500 $0.081 \rightarrow 0.032$ mean absolute error, GSM8K $0.026 \rightarrow 0.018$), and on BoolQ, where $C = 2$ makes the binary law exact by construction, it recovers the same answer, 0.079 — the control condition for the estimator. [C4] The exception is MMLU, where it does *not* beat the binary law (0.079 against 0.061); we have no account of why and do not claim the blind estimator is validated there.

VI. EVALUATION UNDER A SYMBOLIC ADVERSARY

A. Setup

The frozen roster is 7 open-weight models spanning 3B to 32B from seven lineages, plus a separately reported 2-model weak-tier arm never pooled with it. Of 6 benchmarks, 3 are headline-eligible after excluding cells whose honest accuracy leaves no headroom; each carries 200–500 frozen tasks and 500 bootstrap resamples. Accuracy spans 0.340–0.930, so no cell is at ceiling or floor.

This section reports the symbolic adversary—a bloc that emits a designated wrong answer with a specified coherence; the prompt-injected adversaries follow below, and the two regimes disagree in ways that matter.

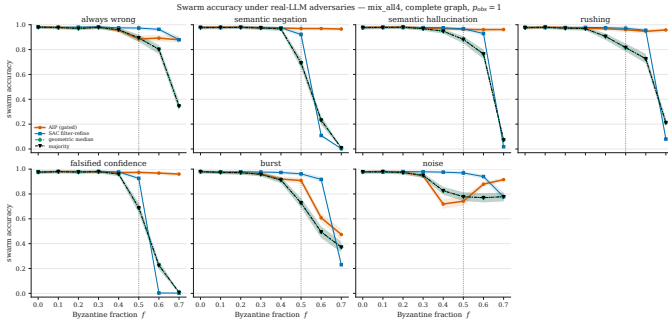


Fig. 1. Swarm accuracy against Byzantine fraction. AIP-gated degrades where discard-based baselines collapse; the benchmarks on which it does *not* degrade are the exception and are named in the text. [H1, H2’]’]

B. The headline, and where it does not hold

Against the symbolic coherent adversary AIP-gated holds 0.976–0.984 on GSM8K across the whole Byzantine range, a spread of 0.008, and 0.877–0.920 on MMLU, a spread of 0.043. Both spreads are below their cells’ measurement floors, so what the data supports is that no change was detected—not that accuracy is provably constant. Every deployable discard baseline reads 0.000 at $f = 0.7$ in the same configuration (Figure 1).

Refuted: accuracy is not flat in f in general. On MATH-500 the same mechanism spans 0.000–0.477, a spread of 0.477, falling to 0.000 at $f = 0.7$ from 0.477 at $f = 0$. Under real-LLM adversaries the fall is present on four of six benchmarks: -0.606 on BoolQ, -0.406 on MedQA, -0.338 on MMLU and -0.307 on ARC, all at least three times the floor. Only GSM8K (-0.014) and MATH-500 (-0.086) are undetectable. The claim this project set out to make was invariance in f ; the claim the data supports is invariance on two benchmarks under one adversary, and a large, ordered degradation elsewhere. [H1]

C. The discard family: three separate statements

“Every discard baseline reaches zero” is true only under conditions that must be stated, and false under two of them. (i) **Symbolic adversary, deployable methods:** true, and exactly — 0.000 on all 6 benchmarks at $f = 0.7$. (ii) **Oracle methods, either regime:** false — Krum, handed the true Byzantine count, holds 0.815 on BoolQ and 0.745 on MedQA, and multi-Krum beats AIP under the real-LLM adversary on both. (iii) **Real-LLM adversary:** false — nothing reaches zero, deployable discard spans 0.002 to 0.600, and AIP separates on four of six (+0.857 GSM8K, +0.528 MATH-500, +0.273 ARC, +0.200 MMLU), is undetectable on MedQA (+0.020), and **loses to plain majority vote on BoolQ by -0.295** . [H2’, H2’’, H2’’’]

BoolQ is the informative failure: with $C = 2$ every wrong answer is the same wrong answer, so adversarial coherence is unity by construction and carries no information. The gate fires on a channel whose sign it cannot use, and majority vote does better by ignoring it — the coordinatability argument arriving as a measured cost.

D. The invertibility spectrum

Inversion gain tracks measured channel coherence, the mechanism’s central prediction: negation reaches $\hat{q} = 0.674$ and gains +0.044, hallucination reaches 0.066 and gains +0.032. **Honest negative result:** semantic hallucination is fluent and highly effective as an attack and AIP can do nothing about it — with no coherent channel there is no sign to recover and the mechanism degrades to discard level. An adversary that simply refuses to coordinate is a complete defence against inversion, at the cost of being a weaker attack. [I2]

E. Why the gate is necessary

Ablating inversion — replacing the three-way decision with trust-or-discard — isolates what it contributes: at $f = 0.7$ the gain is 0.645, at $f = 0$ the cost is 0.001, far below the floor. Inversion is close to free when idle and large when engaged, which is the asymmetry the mechanism is built for. [I1, I2] The gate keeps the first half true: across every configuration tested it inverts at most 0.0% of honest channels while inverting 80.6% of adversarial channels at $f \geq 0.5$. [G1]

F. Two results that did not go our way

The gain is not independent of honest competence. We predicted inversion gain is a property of the adversarial channel, and that a weaker swarm would gain *more*, having more honest error to correct. On a matched benchmark set the weak-tier arm reads 0.470 against the roster’s 0.645, a difference of +0.175 — above the floor, lower on all three benchmarks individually, and opposite to the prediction. The comparison that originally supported the claim pooled the weak arm’s six benchmarks (0.547) against the roster’s three; correcting it reverses the conclusion. [I3]

Whether the gain scales with swarm size is undetectable. At $N = 10, 20$ and 50 the gain reads 0.643, 0.637 and 0.655 — a range of 0.018 against a 0.099 floor. *Neither* growth nor invariance in N is supported; the experiment was underpowered to separate them, and we report it as underpowered rather than as evidence for the flat reading its point estimates suggest. [N1]

G. Real-LLM adversaries

The symbolic adversary above is a control: its coherence is set, not measured. Prompt-injected models are why the discard family does not collapse to zero and why AIP loses on BoolQ.

Non-stationary. A burst adversary alternates coherent and noisy epochs. Pooled history averages the two into a channel clearing neither threshold and the pooled gate falls to 0.766 on GSM8K; sliding-window statistics recover 0.975 at $W = 20$, a gain of +0.210, more than seven times the floor. On a stationary control attack windowing changes accuracy by at most +0.200 — below floor, so *undetectable* rather than zero cost. [B1]

Sleepers. Refuted, and in the opposite direction. We predicted a sleeper would defeat reputation decay while AIP’s windowed statistics reclassified it. Reputation decay degrades *least* of the memory-carrying defenders (0.031) and AIP-gated

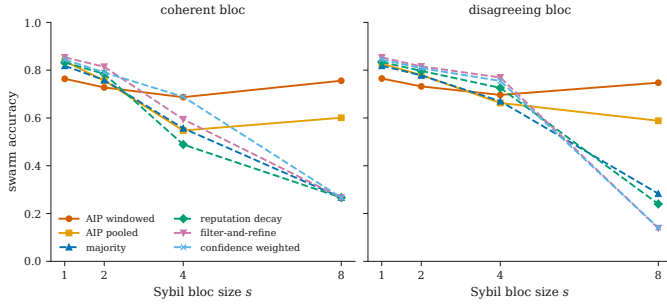


Fig. 2. Accuracy against Sybil bloc size, coherent and disagreeing blocs at a shared scale. The two panels are the test of the amplification explanation, and they do not separate. [E2a, E2b]

most (0.136, windowed 0.121). The spread, 0.105, barely clears the 0.099 floor — marginal, and reported as marginal — but its direction is unambiguous. A defence estimating a channel’s sign per question is more disturbed by a regime change than one averaging a scalar over rounds. [E1]

Sybil blocs. At the largest bloc tested, $s = 8$ — a plurality of the swarm — windowed AIP holds 0.756 against a coherent bloc where the best non-AIP defender reads 0.267, a separation of more than four times the floor (Figure 2). [E2a] *But the mechanism we proposed is refuted:* we predicted the robustness comes from bloc coherence being amplified into a stronger inversion signal, which predicts a gap between a coherent bloc and one whose members disagree. There is none — 0.756 against 0.748, far below floor. The robustness is real; the explanation is not. What the data is consistent with is simpler: receiver-anchored statistics do not care how many identities a bloc presents, because they are not counting. [E2b]

H. What the defence costs

A defence that has to be argued for on robustness grounds should also be priced. Two costs, kept separate because they differ by three orders of magnitude.

Aggregation cost is measured by timing each rule over the cached broadcasts, in microseconds per receiver-decision — one aggregation per question per agent, which is what a deployed agent pays. It is not recoverable from our phase logs, which record one timestamp per benchmark-composition block covering all fourteen rules at once, so it was measured directly rather than apportioned from a block average. *Inference* cost is read from the cache and is identical for every rule: all of them consume the same broadcasts.

The ordering is the useful part. AIP-gated is more expensive than majority vote by a factor of about twenty-five and cheaper than the geometric median by a factor of eight, and the *gate itself* — the binomial test, the ceiling comparison and the signed reweighting — accounts for 3.6% of its cost. Almost all of what AIP spends is spent on maintaining channel statistics, which the inversion-free ablation also pays. Whatever the case against threshold-gated inversion is, it is not compute. [E3]

TABLE II

COST OF DEFENCE AT $f = 0.5$, MICROSECONDS PER RECEIVER-DECISION, MEAN OVER THE HEADLINE BENCHMARKS. THE GATE IS NEARLY FREE: AIP-GATED COSTS $+2.7 \mu s$ OVER ITS OWN INVERSION-FREE ABILATION, OR 3.6%. NOTHING HERE IS THE BOTTLENECK — ONE COMPLETION AVERAGES 634 TOKENS, WHICH AT A GENEROUS 100 TOKENS/S IS ABOUT $6598\times$ THE SLOWEST RULE IN THE TABLE.

Rule	μs / decision
majority vote	3.2
Krum (<i>oracle</i>)	21.2
trimmed mean (<i>oracle</i>)	61.8
coordinate median	83.8
AIP, inversion ablated	74.4
AIP-gated	77.1
Multi-Krum (<i>oracle</i>)	87.3
geometric median	733.1

I. Methodological controls

Normalization parity. The unweighted and discard family is exactly parity-invariant (maximum 0.0000 across every cell), the sanity check that the audit instrument is not manufacturing its own effect; every weighted method responds, so self-vote share is a real confound. [M2a] *Refuted:* AIP does exploit it at the tie. Away from $f = 0.5$ the effect is undetectable (largest 0.135); at $f = 0.5$ it averages 0.212 and reaches 0.840, so with the swarm split evenly the normalisation of the adversary’s own vote decides the outcome. Every number we state at the tie, the evasion anchor included, is measured in the matched-parity condition. The pooled average 0.0283 hides this and should not be quoted alone. [M2b]

Minimax regret over unknown f , deployable methods only: AIP-gated is best on GSM8K (0.004 against 0.963) and MMLU (0.038 against 0.902), and *not* on MATH-500 (0.523 against 0.438). [M1]

VII. THE GATE-AWARE ADVERSARY

Every adversary so far has been oblivious to the defence. This one is not. It is given the published calibration constant — the honest-coincidence ceiling for the answer space — and it is asked to do the obvious thing: coordinate as much as it can without crossing it.

A. Construction

A bloc coordinating with probability p and otherwise picking a wrong answer uniformly realises $q(p) = p^2 + (1 - p)^2 / (C - 1)$, so sweeping p sweeps q and the adversary picks the p whose q does most damage while staying below the ceiling. The map is not monotone — q falls as p rises from zero, reaches an interior minimum, then rises — which is why the adversary has a genuine optimum rather than a corner solution, and why Figure 3 plots points and not a curve.

B. The anchor is $f = 0.5$

The anchor is the exact tie, and the choice has to be defended because the more dramatic numbers live elsewhere. At $f = 0.7$ the Byzantine bloc is an outright majority and *every* coherence level, independent lying included, drives accuracy toward zero;

a collapse there cannot be attributed to the gate, and an evasion effect is not separable from numerical dominance. An earlier version of this work reported the $f = 0.7$ attacker gain; it is withdrawn. At $f = 0.5$ the outcome turns on whether the gate fires — and, at that point only, on the normalisation of the adversary’s own vote (Section VI), so the sweep runs in the matched-parity condition and the confound is controlled rather than assumed away.

C. Result

The gate-aware adversary defeats AIP. At its optimum, $\hat{q} = 0.280$, AIP-gated falls to 0.082 on MedQA against 0.878 under a fully coherent adversary—an attacker gain of +0.796, eight times the floor. On MMLU the same optimum yields 0.374 against 0.903, a gain of +0.529. [G1]

The number that makes this more than an engineering setback is the optimum’s location. The chance coherence for a four-option space is $1/(C-1) = 0.333$. The attacker’s optimum sits at 0.280: **below chance**. At its most damaging the adversary is not merely quieter than the calibrated ceiling—it is *less* coherent than honest agents failing independently. Raising, lowering, smoothing or randomising a threshold cannot help, because there is no threshold on this axis that separates the attacker from honest disagreement. Section VIII makes that argument precisely.

MATH-500 behaves differently and the difference is instructive. There the attacker’s global optimum is a *fully coherent* channel, not an evading one: the open-text ceiling of the answer space is high enough relative to achievable coherence that the gate is near-inert, and the adversary has nothing to evade. Its best below-ceiling attack yields 0.366, which is *worse* for the attacker than coherence—a gain of -0.366 . A low ceiling is not a structural defence. [L1]

D. Pre-registered mitigations

Three mitigations were specified, with success criteria committed to version control before any of them ran.

M1, soft gating. Replace the hard binomial threshold with a logistic function of the coherence ratio, so that a channel just under the ceiling is partially inverted rather than fully trusted. Within the evasion band—the cells below the ceiling, which is where the residual attack lives—mean accuracy rises from 0.463 to 0.623 and the worst case from 0.082 to 0.200. This is a real improvement of about one and a half times the floor.

Over the *whole* attack space at the anchor, however, the improvement is 0.457 to 0.557—exactly at the floor. Only the band-restricted statement is defensible, and M1 fails its pre-registered success criterion regardless: it does not bring the worst case anywhere near the committed target. [D1]

M2, threshold randomisation. Refuted, and undetectably so. Randomising the threshold within a relative window was intended to deny the adversary a fixed target. It reads 0.444 against the hard gate’s 0.457 over the whole space and 0.474 against 0.463 within the band. Both differences are far below the floor. Randomising a statistic that carries no signal buys no discrimination—in either direction. An earlier reading of

this experiment called M2 “actively harmful” on the strength of a difference that was itself below the floor; that reading was wrong in the same way, and is withdrawn. [D2]

M3, both together. Band mean 0.638, worst case 0.278. Not separable from M1 alone.

Both M1 and M2 were initially inert, returning results identical to the unmitigated baseline, and those identical results were briefly read as a finding. They were wiring bugs, caught by probe tests that assert variant activity before any verdict is read, and two published verdicts were retracted; Section IX gives the detail and why it belongs in the paper. [R1]

VIII. LIMITS OF COHERENCE-BASED GATING

Section VII showed an adversary whose optimum coherence is below chance. This section argues that no amount of threshold engineering recovers from that, and identifies what a successor defence would have to bring instead.

A. Proposition 1, informally

Fix a question with a closed answer space of size C . AIP’s gate is a decision rule computed from a channel’s per-question coherence q , the probability that two agents on the channel, both wrong, are wrong the same way. Honest agents failing independently collide at the chance rate $q_{\text{ind}} = 1/(C-1)$.

Let $\mathcal{A}$ be an adversarial channel with $q_{\mathcal{A}} \leq q_{\text{ind}}$, and $\mathcal{H}$ a population of honest agents failing independently on the same space. Then for any decision rule computed solely from per-question coherence, the sum of its false-positive and false-negative rates is at least $1 - d_{TV}$, where d_{TV} is the total variation distance between the statistic’s distributions under the two hypotheses, and $d_{TV} \rightarrow 0$ as $q_{\mathcal{A}} \rightarrow q_{\text{ind}}$.

The argument is short. Collision counts among jointly-dissenting peers are a sufficient statistic for q under the exchangeability the estimator already assumes; by the Neyman–Pearson lemma the best achievable separation between two hypotheses on a common statistic is governed by the total variation distance between its distributions; the count is Binomial(n, q), so d_{TV} is continuous in q and vanishes as the two parameters meet. The bound therefore constrains *every* rule measurable with respect to that statistic, not the particular threshold we chose. The formal setup, the exchangeability condition in full and the finite- n statement are in the artifact’s technical note; the substance is that d_{TV} degrades as $\sqrt{n}$ in the number of jointly-dissenting observations, which is bounded by the swarm size — so the escape route is longitudinal rather than per-question — and that at $q_{\mathcal{A}} = q_{\text{ind}}$ exactly, $d_{TV} = 0$ for every n .

This is why M1 helps a little and M2 not at all: soft gating recovers accuracy in the band below the *calibrated ceiling* but still above chance, where the statistic carries signal, while at and below chance there is nothing for any function of q to find and randomising the threshold only adds variance.

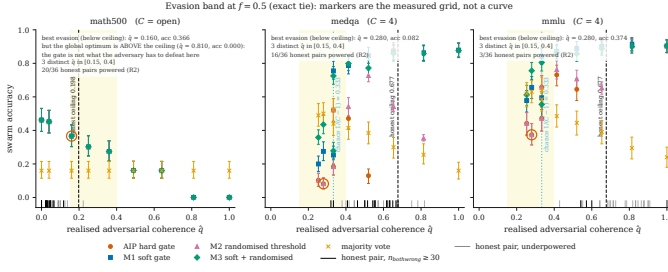


Fig. 3. The evasion band at the anchor, $f = 0.5$ (exact tie); markers are the measured grid; nothing is interpolated between them. The dashed line is the calibrated honest ceiling the gate fires against, the dotted line is the chance coherence $1/(C - 1)$ of independent wrong answers, and the rug shows the per-pair honest coherence with underpowered pairs ($n_{\text{both wrong}} < 30$, rule R2) drawn in grey. On MedQA the attacker’s optimum sits below both lines; on MATH-500 the gate is near-inert and the attacker’s optimum is a fully coherent channel, so there is nothing there to evade.

B. What the measurement can and cannot say

The anchor is a single grid point and we are explicit about its resolution: the parameterisation reaches only 3 distinct coherence values inside $[0.15, 0.40]$ and cannot express a channel less coherent than 0.253, the interior minimum of $q(p)$. The optimum is *measured* at a grid point, not interpolated; we claim nothing about its shape between points, and whether a differently constructed adversary could reach lower q is untested.

C. Inversion and evasion are one law, read from two answer spaces

The mechanism’s success and its defeat are not two findings. They are the same inequality evaluated on two answer spaces.

The inequality is whether the adversary’s coherence exceeds the honest receiver-conditioned coincidence on that space. On open-numeric answers under negation it does — $\hat{q} = 0.674$ against an honest floor of 0.597 — so there is a separable channel, the gate fires, and inversion converts the adversary into a contributor. Every benchmark where AIP separates from the discard family is on this side of the line. On four-way multiple choice it does not: 0.285 against 0.535. The adversary is *more* coordinated in absolute terms than on open text and still carries no channel, because the honest agents are more coordinated too. [G2’]

The gate-aware adversary is what happens when an adversary chooses the second side of that line deliberately: its optimum 0.280 sits below the honest floor and below chance 0.333. It is not evading a threshold, it is moving into the regime where no threshold on this statistic has anything to work with — exactly the regime Proposition 1 describes. So the practitioner’s question is not “does inversion work” but “does this answer space put my adversary above the honest floor”; on a binary space the answer is no by construction, which is why majority vote beats AIP on BoolQ. The same law predicts the win and the loss. [D3]

D. Two escape classes, both untested

The proposition constrains defences whose statistic is per-question channel coherence. It says nothing about defences drawing on information that statistic does not contain. Two such classes are identifiable, and both are stated here as **hypotheses that have not been evaluated against the gate-aware adversary**.

External information. Identity verification, stake, or a ground-truth anchor breaks the symmetry the proposition relies on by supplying evidence from outside the channel. This is precisely the gap AgentChain [14] names as unsolved when it admits that its guarantees stop at $f \geq N/2$ (Section II). Our contribution here is not to close that gap but to give a reason to think a coherence-based defence cannot: the information required is not present in the statistic, so no better estimator of the statistic will supply it.

Longitudinal statistics. Cross-task reputation accumulated over rounds is not a function of per-question coherence and is therefore outside the proposition’s scope. Our own sleeper experiment is suggestive in this direction and against our prior expectation: reputation decay degraded least of the memory-carrying defenders, at 0.031 against AIP-gated’s 0.136, a spread of 0.105 that only just clears the 0.099 floor. That is a marginal result pointing at an untested hypothesis, and it is worth exactly that much. A longitudinal defence has not been run against the gate-aware adversary and we do not claim it survives.

E. What this means for the mechanism

Sections VI–VIII describe a defence with a characterised operating envelope rather than a solution: close to free when idle, large when the adversary is coherent and the answer space open, useless when the space is binary, and defeated by an adversary that reads its calibration constant and sits below it. That last is a property of the statistic, not a bug to be patched.

IX. SCOPE AND LIMITATIONS

This section collects every claim this study set out to make and did not, every claim it makes only under conditions, and the instrumentation failures that produced two retracted verdicts. It is placed before reproducibility rather than after the conclusion because the refutations are results.

A. The measurement floor, and what it forbids

The floor is 0.066 on GSM8K and 0.099 elsewhere (§I). Applying it consistently cost this paper three findings — an N -scaling result, a mitigation verdict and a Sybil mechanism — which is the point of fixing the rule before seeing which way it cuts.

B. Refuted: claims we set out to make and cannot

Two refutations are given in prose because they revise the paper’s own headline and because a table cannot carry the qualification each needs. The remaining eight are tabulated in Table III; tabulating them raises rather than lowers their visibility, and every row carries its magnitude and its floor.

Accuracy is not flat in the Byzantine fraction. This is the claim the study was built to make, and it does not survive. Under the real-LLM coherent adversary AIP-gated falls by -0.606 on BoolQ, -0.406 on MedQA, -0.338 on MMLU and -0.307 on ARC between $f = 0$ and $f = 0.7$, all at least three times the floor. GSM8K (-0.014) and MATH-500 (-0.086) are undetectable, so flatness is there neither established nor refuted — which is a weaker statement than either “flat” or “not flat”, and is the one the data supports. Under the symbolic adversary MATH-500 spans 0.477. What survives is invariance on specific benchmarks under a specific adversary, and we state both names wherever it appears. [H1]

An earlier evasion anchor is withdrawn, and a deterrence result with it. The attacker gain this project first reported was read off the $f = 0.7$ row, where the Byzantine bloc is an outright majority and every coherence level including independent lying drives accuracy to zero; that row cannot separate an evasion effect from numerical dominance and is not an admissible anchor. The anchored value is $+0.796$ at the tie. Separately, an adaptive-adversary deterrence experiment is absent from this paper rather than reported with a caveat: the condition for including it, fixed before the audit ran, was that the coordinatability claim survive re-verification, and it did not. Its data is in the artifact. [G1, B2]

C. Undetectable at this scale

Whether inversion gain depends on swarm size. At $N = 10, 20$ and 50 the gain reads $0.643, 0.637$ and 0.655 : a range of 0.018 against a 0.099 floor. Neither growth nor invariance in N is supported, and the figure caption says so rather than asserting the flat reading its point estimates suggest. [N1]

D. Claims we explicitly do not make

Accuracy is not monotone increasing in f , as an early simulation predicted: only the ungated variant shows it, and that variant is catastrophic at low f . [U1]

No generality beyond this population. 7 models (3–32B) plus a 2-model weak arm, 6 benchmarks, 200–500 tasks each, one adversary family per attack class, one hardware regime. The honest-coincidence ceilings must be re-measured for any other population. [U2]

No claim that the escape classes survive. Neither the external-information nor the longitudinal defence of Section VIII has been evaluated against the gate-aware adversary. [U3] And blind bootstrapping is not established everywhere: the multiclass law does not beat the binary law on MMLU (0.079 against 0.061).

E. Where the mechanism fails by design

- **Incoherent adversaries.** No coherent channel, nothing to invert; a discard method is the correct choice. [I2]
- **Binary answer spaces.** Coherence is unity by construction and carries no information; majority vote beats AIP on BoolQ by -0.295 .
- **Non-stationary adversaries.** Pooled statistics are defeated; windowing mitigates but does not eliminate. [B1]

TABLE III

THE OTHER EIGHT REFUTATIONS. EACH IS A CLAIM THIS STUDY SET OUT TO CONFIRM AND ITS OWN DATA REFUSED. MAGNITUDES ARE AGAINST THE BENCHMARK’S MEASUREMENT FLOOR (0.066 ON GSM8K, 0.099 ELSEWHERE).

Claim	What the data says
[L1] A low honest- q ceiling is a structural defence	MATH-500 has the lowest ceiling and the gate is near-inert there: the gate-aware adversary’s best strategy is <i>full coherence</i> , not evasion. A low ceiling means the gate rarely fires, not that the adversary rarely succeeds.
[I3] Inversion gain is independent of honest competence	Matched benchmark set: weak arm 0.470 against the roster’s 0.645 , a gap of $+0.175$, lower on all three individually. The supporting comparison had read six benchmarks against three.
[S1] Maximising effective swarm size is the composition recipe	N_{eff} tracks mean roster accuracy at -0.934 — it restates roster weakness — and swarm accuracy at -0.506 , the wrong sign. No composition beats its own best single member detectably.
[M2b] AIP does not exploit self-vote share	Undetectable away from the tie (largest 0.135); at $f = 0.5$ it averages 0.212 and reaches 0.840 . Every number quoted at the tie must name its parity condition.
[E2b] Sybil robustness comes from coherence amplification	AIP is as robust to a disagreeing bloc (0.748) as to a coherent one (0.756). The robustness holds; the explanation does not.
[E1] A sleeper defeats reputation decay while AIP reclassifies it	Reversed. Reputation decay drops least (0.031), AIP most (0.136); spread 0.105 against a 0.099 floor, so marginal, and reported as marginal.
[G2’] Coordinatability collapses on closed label spaces	The direction is inverted: coherence is <i>highest</i> there (0.559 binary, 0.285 multiple choice, against 0.104 open text). What collapses is the margin over honest agreement.
[D2] Threshold randomisation helps	-0.013 over the attack space and $+0.010$ within the band, both far below floor: undetectable in <i>either</i> direction. An earlier reading called it “actively harmful” on a difference that was itself below floor.

- **Partial observability.** Accuracy falls from 0.980 at $p_{\text{obs}} = 1$ to 0.917 at $p_{\text{obs}} = 0.5$ on GSM8K at $f = 0.6$. This, not the Byzantine fraction, is the binding practical constraint.
- **Gate-aware adversaries.** Section VII, and the reason is structural rather than incidental (Section VIII).

F. Declared data gaps

The binary answer-space class carries no calibrated honest receiver-conditioned coincidence, so BoolQ’s ceiling is a placeholder rather than a measurement and no margin-over-honest claim is made for it. This is recorded as a gap rather than absorbed into the results. [X1]

G. Instrumentation failures and retracted verdicts

Two mitigation variants were initially inert and returned results identical to the unmitigated baseline. Those identical results were read as a finding—that the mitigations do not matter—and published internally as such. They were bugs: M1

fell through to the hard gate for every adversarial channel, and M2’s jitter reached only a fixed-ceiling ablation rather than the default rule. Both verdicts were retracted.

They were caught by probe tests that assert variant activity before any verdict is read. We report this beyond mere disclosure: a mitigation study in which a true null and a wiring bug produce the same observation cannot tell them apart from the output alone, and an artifact that detects its own instrumentation failures is stronger evidence than one that cannot. A related error accounts for three refutations above — comparisons across mismatched bases (six benchmarks against three, an upper-bound estimator against a measured one, an $f = 0.7$ row against an $f = 0.5$ claim) where the mismatch and not the effect produced the result. The audit that found them runs in the build. [R1]

X. REPRODUCIBILITY AND CONCLUSIONS

A. Artifact

All generation happens once and is written to parquet; aggregation, baselines, mitigations and analysis read that cache and never invoke a model. That is what makes a sweep of this size affordable on one device, and what makes the offline attacks exact — `falsified_confidence` and `burst` are transforms of a cached adversarial run, byte-identical to their control outside the one channel each manipulates. Batch-invariant attention is enabled for every cache-generating run; without it identical batches return logprobs differing at the sixth decimal, enough to flip a near-tie argmax. Seeds derive from a SHA-256 of the canonical configuration, task identifiers are frozen and committed, and every run writes a manifest with its configuration hash, git commit, seeds, GPU and package versions. All results come from a single hardware regime in `bfloat16`; an earlier regime with a different model population is not mixed in, for any claim.

A completion budget is a measurement instrument, and every benchmark here commits its answer at the *end* of the completion, so a truncated generation is the one missing its answer. Auditing by `finish_reason` rather than by parse rate — 100% throughout, and not a health metric — found 1036 of 13500 generations at the cap and 944 whose answer the extractor had read out of unfinished working (0.351 on MATH-500). We raised every budget to 3072 tokens and regenerated those rows alone at the same prompts and seeds, checking per cell that untouched rows came back byte-identical. Ledger entry X3 itemises what moved; every number here is post-repair.

Every number here is a macro emitted from the result parquets — a hand-typed decimal fails the build — and the ledger, the section-to-claim map and the manuscript are checked against each other mechanically: a cited claim must exist in the ledger, a ledger entry must map to a section, no claim may remain unaudited against the floor, and every figure must name the script that produced it. Those checks caught four constants from an earlier regime surviving inside macros, and seven stale figures from it. The release contains the registry, the frozen task lists, the cache, every result

parquet, the generators, the ledger, the pending-claim audit, the technical note supporting Proposition 1, and the probe tests of Section IX. [Archive DOI pending de-anonymisation.]

B. Conclusions

The 50% ceiling is a property of discarding, not of the problem. Correlated error caps what honest pooling can buy — established for evaluation panels [3], [4], confirmed here in an adversarial swarm and extended to the replicated-model case — which leaves adversarial coherence as the only informative structure a swarm has that its honest members do not already share. Inversion is worth 0.645 at $f = 0.7$ and costs 0.001, below the floor, at $f = 0$. Coherence alone cannot license it, because honest agents cohere at 0.547 against chance 0.333 on the powered four-way multiple-choice pairs; the gate is calibrated per answer space and anchored on the receiver’s own answer, which lets it work past the point where consensus-based estimators have converged on the lie.

The limit is not reachable by threshold engineering. An adversary that reads the published ceiling and coordinates below it drives AIP to 0.082 at the tie, with an optimum at 0.280 — below the chance coherence 0.333 of honest independent failure. Escaping that needs information the channel does not contain: external verification, which the systems literature names as open, or longitudinal statistics, which our sleeper result gestures toward and which we have not tested. Both are hypotheses, and we state them as such.

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